Jianting Zhang

Ph.D. Candidate, Department of Computer Science, Purdue University

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Research Interests

My research lies at the intersection of distributed systems, security, and AI. I study how to build scalable, high-performance distributed systems that remain resilient to failures and adversarial behavior, with interests in trustworthy AI-agent systems, fault-tolerant distributed AI training, Byzantine fault-tolerant state machine replication, and blockchain security, including front-running and MEV.

Selected Research Contributions

- **Trustworthy AI-Agent Systems**: *TrustWeir*: Sybil-resilient payment authorization for permissionless AI-agent markets with bounded financial exposure.
- **Resilient BFT Systems**: Scalable, low-latency, and resilient BFT protocols, including *Beluga* (adopted by Sui), *Lifefin*, and *Pufferfish*.
- **Scalable Blockchain Infrastructure**: Deconstructed-SMR sharding achieving 180,000+ transactions per second on 500 geo-distributed nodes; adopted by Supra and published at VLDB 2025.
- **Blockchain Security**: Front-running/MEV attacks for sharded and DAG-based blockchains, published at NDSS 2024 and ACSAC 2025.

Education

- **Purdue University** — West Lafayette, IN, USA
  Ph.D., Computer Science, advised by Aniket Kate — *August 2022–May 2027 (expected)*
- **Sun Yat-sen University** — Guangzhou, China
  M.S., Computer Science and Technology, advised by Wuhui Chen and Zibin Zheng — *August 2019–June 2022*
- **Sun Yat-sen University** — Guangzhou, China
  B.S., Computer Science and Technology — *August 2015–June 2019*

Honors and Awards

- Bilsland Dissertation Fellowship, Purdue University, 2026–2027
- Graduate Student with Honors, Sun Yat-sen University, 2022
- ICPP Best Paper Award Runner-up, 2020
- National Scholarship (The highest-level national award in China), 2020

Industry Research Experience

- **Mysten Labs**: Research Intern, Remote, May 2025–August 2025
- **Alibaba DAMO Academy**: Research Intern, Beijing, May 2020–August 2020

Selected Publications

Peer-Reviewed Conference Publications

- **NASDAQ: Normalized Observation Space Dynamics-Augmented Q-Learning**
  Xinwei Liu, Junyuan Liang, Zicong Hong, **Jianting Zhang**, Wuhui Chen
  *The 40th Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026*
- **Five Minutes of DDoS Brings down Tor: DDoS Attacks on the Tor Directory Protocol and Mitigations**
  Zhongtang Luo, **Jianting Zhang**, Akshat Neerati, Aniket Kate
  *European Conference on Computer Systems (**EuroSys**), 2026*
- **Optimal Sharding for Scalable Blockchains with Deconstructed SMR**
  Jianting Zhang, Zhongtang Luo, Raghavendra Ramesh, Aniket Kate
  51st International Conference on Very Large Data Bases (**VLDB**), 2025
  Adopted by Supra Blockchain

- **No Fish Is Too Big for Flash Boys! Frontrunning on DAG-based Blockchains**
  Jianting Zhang, Aniket Kate
  *41st Annual Computer Security Applications Conference (**ACSAC**), 2025*
- **Front-running Attack in Sharded Blockchains and Fair Cross-shard Consensus**
  Jianting Zhang, Wuhui Chen, Sifu Luo, Tiantian Gong, Zicong Hong, Aniket Kate
  *31st Network and Distributed System Security Symposium (**NDSS**), 2024*
- **Porygon: Scaling Blockchain via 3D Parallelism**
  Wuhui Chen, Ding Xia, Zhongteng Cai, Hong-Ning Dai, **Jianting Zhang**, Zicong Hong, Zibin Zheng
  *40th International Conference on Data Engineering (**ICDE**), 2024*
- **Auncel: Fair Byzantine Consensus Protocol with High Performance**
  Wuhui Chen, Yikai Feng, **Jianting Zhang**, Zhongteng Cai, Hong-Ning Dai, Zibin Zheng
  *43rd IEEE International Conference on Computer Communications (**INFOCOM**), 2024*
- **PROPHET: Conflict-Free Sharding Blockchain via Byzantine-Tolerant Deterministic Ordering**
  Zicong Hong, Song Guo, Enyuan Zhou, **Jianting Zhang**, Wuhui Chen, Jinwen Liang, Jie Zhang, Albert Zomaya
  *42nd IEEE International Conference on Computer Communications (**INFOCOM**), 2023*
- **SkyChain: A Deep Reinforcement Learning-Empowered Dynamic Blockchain Sharding System**
  Jianting Zhang, Zicong Hong, Xiaoyu Qiu, Yufeng Zhan, Song Guo, Wuhui Chen
  49th International Conference on Parallel Processing (**ICPP**), 2020
  Best Paper Runner-up

Peer-Reviewed Journal Publications

- **GenkiDama: Cross-Chain Rollup for Scalable Blockchain Interoperability**
  Yusheng Qin, Desheng Jiang, Zicong Hong, Wuhui Chen, **Jianting Zhang**, Zibin Zheng, Song Guo
  *IEEE Transactions on Dependable and Secure Computing (**TDSC**), 2026*
- **Scaling Blockchain via Dynamic Sharding**
  Jianting Zhang, Zicong Hong, Xiaoyu Qiu, Yufeng Zhan, Song Guo, Wuhui Chen
  *IEEE Transactions on Dependable and Secure Computing (**TDSC**), 2025*
- **RapidSnail: Improve Scalability of Blockchain under High Contention Workload**
  Junyi Wen, Wuhui Chen, **Jianting Zhang**, Xiao Chen, Ting Cai, Hong-Ning Dai, Zibin Zheng
  *IEEE Transactions on Computers (**TC**), 2025*
- **Efficient Execution of Arbitrarily Complex Cross-shard Contracts for Blockchain Sharding**
  Jianting Zhang, Wuhui Chen, Zicong Hong, Gang Xiao, Linlin Du, Zibin Zheng
  *IEEE Transactions on Computers (**TC**), 2024*
- **SPARROW: Expediting Smart Contract Execution for Blockchain Sharding via Inter-shard Caching**
  Junyuan Liang, Peiyuan Yao, Wuhui Chen, Zicong Hong, **Jianting Zhang**, Ting Cai, Min Sun, Zibin Zheng
  *IEEE Transactions on Parallel and Distributed Systems (**TPDS**), 2024*
- **Benzene: Scaling blockchain with cooperation-based sharding**
  Zhongteng Cai, Junyuan Liang, Wuhui Chen, Zicong Hong, Hong-Ning Dai, **Jianting Zhang**, Zibin Zheng
  *IEEE Transactions on Parallel and Distributed Systems (**TPDS**), 2022*

Manuscripts and Preprints

- **TrustWeir: Sybil-Resilient Payment Authorization for AI Agents**
  Jianting Zhang, Zhongtang Luo, Aniket Kate
  Under review
- **Pufferfish: DAG-Guided Speculative Execution for Low-Latency BFT SMR**
  Jianting Zhang, Alberto Sonnino, Lefteris Kokoris Kogias, Aniket Kate
  Under review
- **Beluga: Block Synchronization for BFT Consensus Protocol**
  Jianting Zhang, Lefteris Kokoris Kogias, Tasos Kichidis, Arun Koshy, Mingwei Tian, Ilya Sergey, Alberto Sonnino
  Under review
  Adopted by Sui Blockchain

- **Lifefin: Escaping Mempool Explosions in DAG-based BFT**
  Jianting Zhang, Sen Yang, Alberto Sonnino, Sebastián Loza, Aniket Kate
  Under review

Research Mentoring

- **Purdue University & Sun Yat-sen University**
  Research Mentor *January 2021 – May 2026*
  - **Sebastian Loza, Visiting Undergraduate, University of Engineering and Technology**: Mentored research on robust DAG-based BFT; collaboration resulted in *Lifefin* (under review).
  - **Akshat Neerati, Undergraduate, Purdue University**: Co-mentored research on Tor reliability and consensus; collaboration resulted in a paper at EuroSys 2026.
  - **Yikai Feng, Master's Student, Sun Yat-sen University**: Mentored research on high-performance order-fair consensus; collaboration resulted in a paper at INFOCOM 2024.
  - **Junyi Wen, Ph.D. Student, Sun Yat-sen University**: Mentored research on parallel blockchain execution under high contention; collaboration resulted in a paper in IEEE TC 2025.
  - **Zhaoxian Yang, Master's Student, Sun Yat-sen University**: Mentored research on on-chain/off-chain blockchain optimization; collaboration resulted in a paper at ICSOC 2023.

Teaching Experience

- **Purdue University** West Lafayette, IN
  Teaching Assistant, CS 18200: Foundations of Computer Science *Fall 2024; Spring 2026*
  - **Instruction**: Led weekly Practice, Study, and Observation (PSO) sessions and office hours on discrete mathematics, formal reasoning, and proof techniques.
  - **Assessment**: Graded assignments and examinations and contributed to the design and review of exam questions.

Service

- **Program Committee (PC)**: ACM CCS'26; USENIX Security'27; FC'27
- **Journal Reviewer**: IEEE/ACM ToN, IEEE TSC, IEEE TIFS, ACM TOPS, ACM TOIT
- **Session Chair**: ACSAC 2025

Selected Talks

- **Five Minutes of DDoS Brings down Tor: DDoS Attacks on the Tor Directory Protocol and Mitigations**:
  European Conference on Computer Systems (**EuroSys**), Edinburgh, UK, 2026
- **No Fish Is Too Big for Flash Boys! Frontrunning on DAG-based Blockchains**:
  Annual Computer Security Applications Conference (**ACSAC**), Honolulu, USA, 2025
  The Science of Blockchain Conference (**SBC**), Berkeley, USA, 2025
  The Latest in DeFi Research (**TLDR**), New York City, USA, 2025
- **Front-running Attack in Sharded Blockchains and Fair Cross-shard Consensus**:
  Network and Distributed System Security Symposium (**NDSS**), San Diego, USA, 2024
- **SkyChain: A Deep Reinforcement Learning-Empowered Dynamic Blockchain Sharding System**:
  49th International Conference on Parallel Processing (**ICPP**), Online, 2020